DEPARTMENT OF
CIVIL AND ENVIRONMENTAL
ENGINEERING

SCHOOL OF ENGINEERING

香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

DISC5105 - Autonomous Construction and Robotics

Week 8: Multimodal Data Fusion Lab

Occlusion, plantar pressure, fusion methods, and the Colab lab.

Prof. Yu Yantao

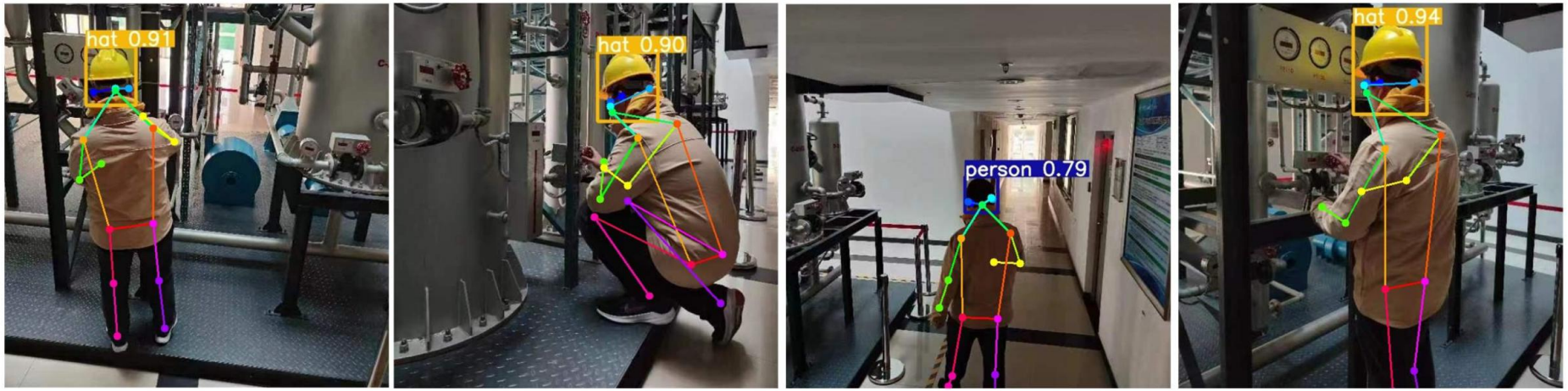
DISC5105 Week 6

The Hong Kong University of Science and Technology (HKUST)



Pose estimation at construction sites

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YOLO-Pose



HRNet

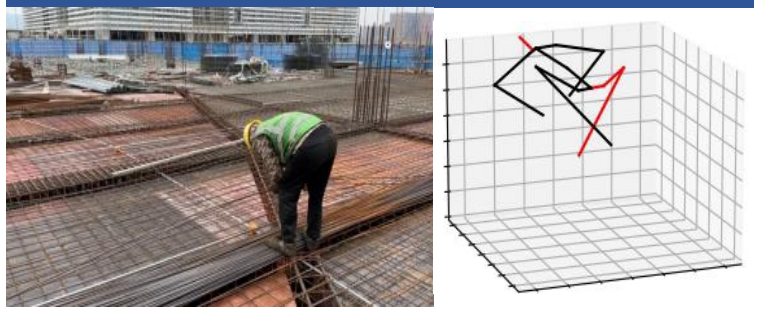
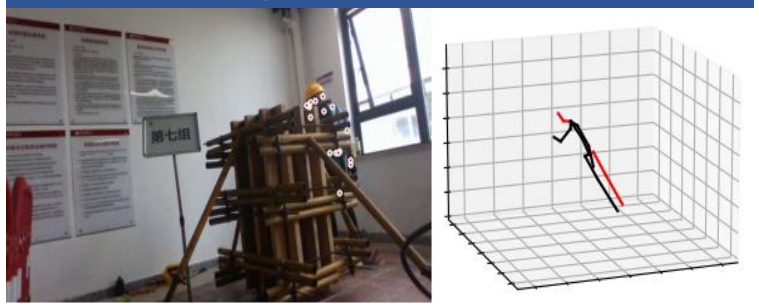
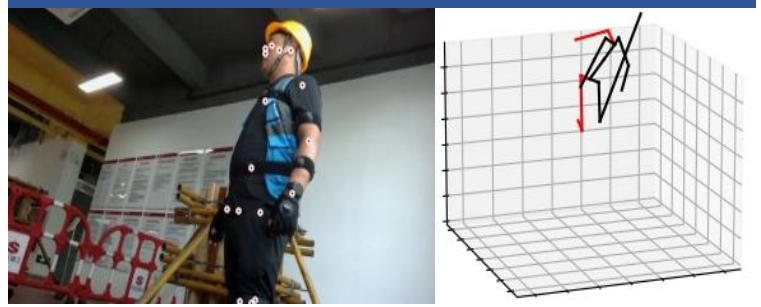
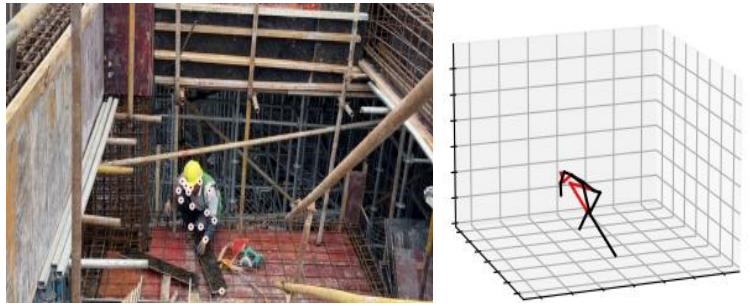
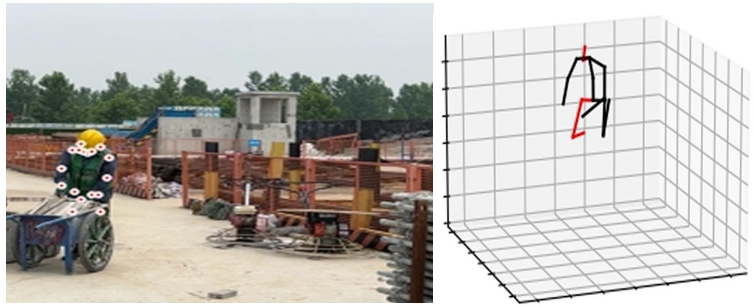
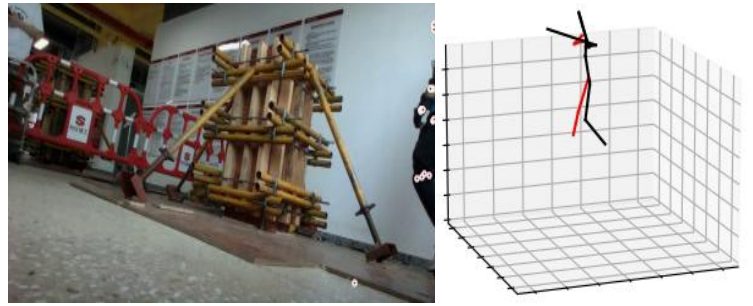
- In real construction sites, fixed CCTV cameras can continuously observe workers.
- Pose-estimation models such as YOLO-Pose and HRNet can detect workers and estimate their body-joint locations.

When a joint is hidden or outside the frame, camera-only pose estimates become uncertain.

The occlusion problem

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- Construction-site cameras are often fixed, while equipment, materials and other workers can block the view.

Self-occlusion	Object occlusion	Outside the camera view
		
		

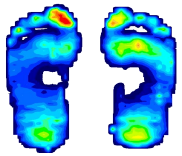
Challenge: incomplete visual evidence

What should we do?

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Camera



Insoles



IMUs



EMG



Other modalities

When a sensor provides incomplete or unreliable observations, we can add a complementary sensor that captures another physical information.

In the previous lecture, we examined several sensors, including vision cameras, plantar-pressure insoles, and IMUs. In the visual occlusion, which sensor can provide complementary information to support human pose estimation?

Which sensor can still help when the camera view is blocked?

Why fuse vision and plantar pressure?

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- Vision-based methods struggle to assess worker postures in occluded construction environments and lack external load information



- Pressure data from smart insoles can assess external load on workers, and pressure distribution can reflect workers' postural characteristics

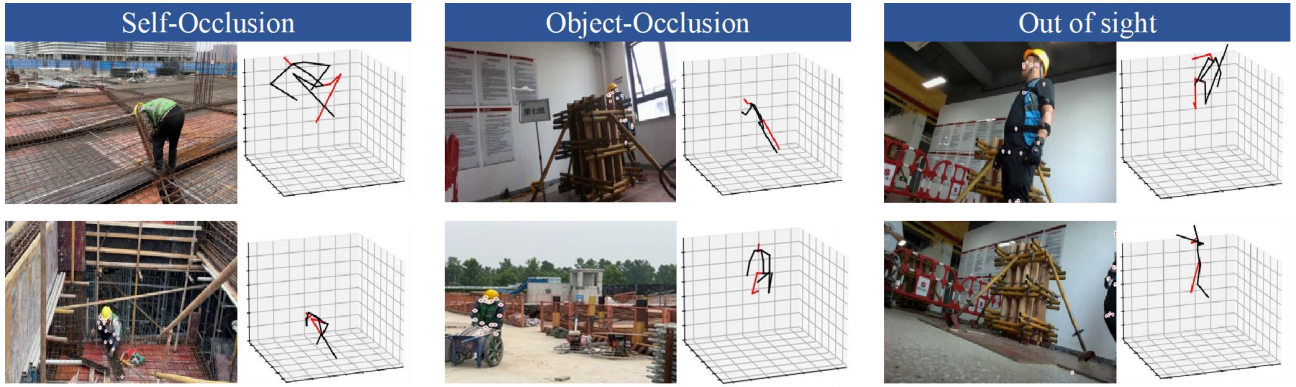


Fig. Examples of 3D pose estimation errors resulting from visually obstructed environments

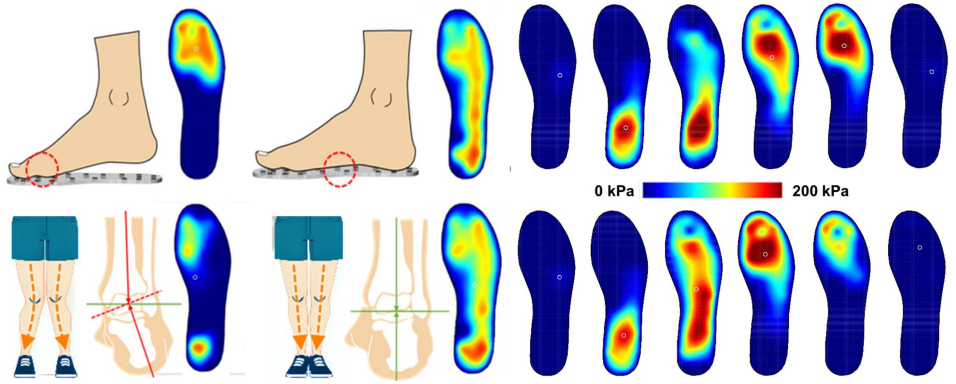
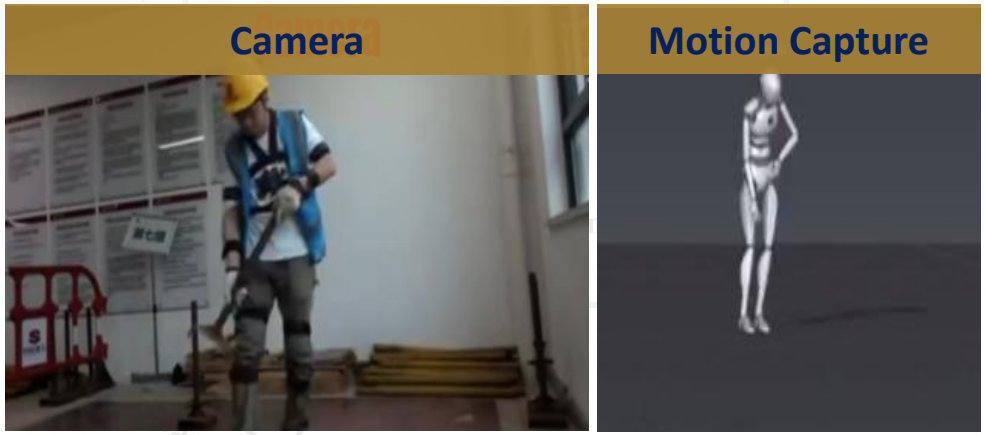


Fig. Examples of insole pressure distributions

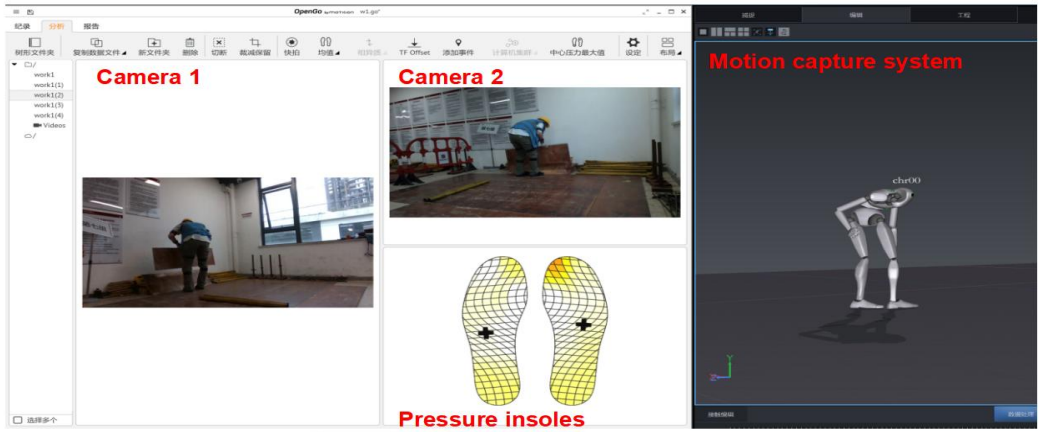
Can we integrate pressure and visual data for worker pose estimation and ergonomic risk assessment in visually occlusion environment?

In our research method, two cameras record the worker from different views. MotionGo smart insoles collect plantar-pressure data. A wearable motion-capture system records the worker's 3D posture and provides ground-truth labels for model training.

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Collect 3D posture data as the ground truth to train our model





- Cameras --> as input image data for multimodal training
- Motiongo smart insoles --> as input pressure data for multimodal training
- Wearable motion capture system --> 3D posture as the ground truth to train our model

Alignment comes before fusion

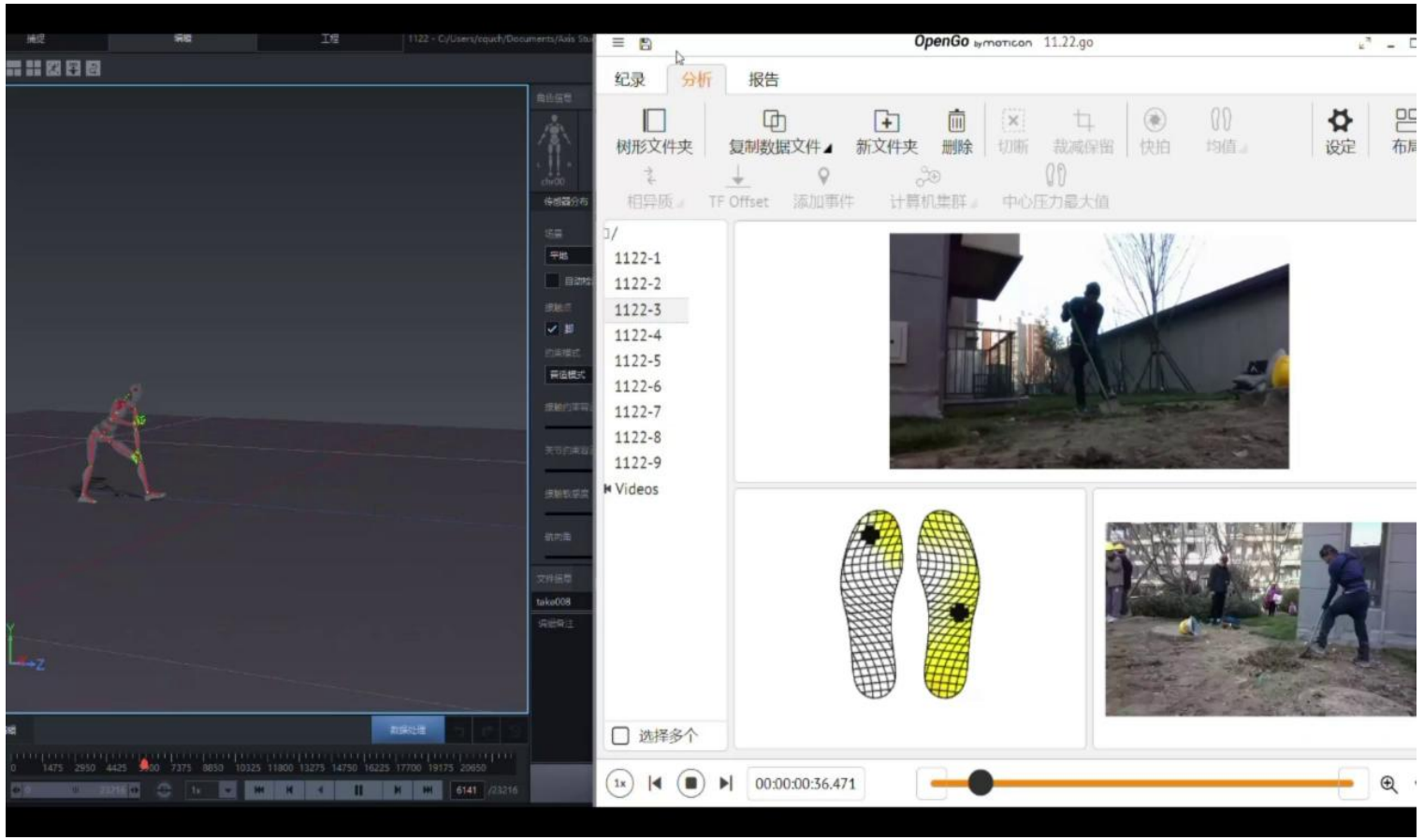
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Why align first?

- Before fusion, match each video frame with foot-pressure data from the same moment.

- Without alignment, the model may learn the wrong link between pressure and pose, making estimation errors.

- As shown on the right, we use a common timestamp across devices to align the data.



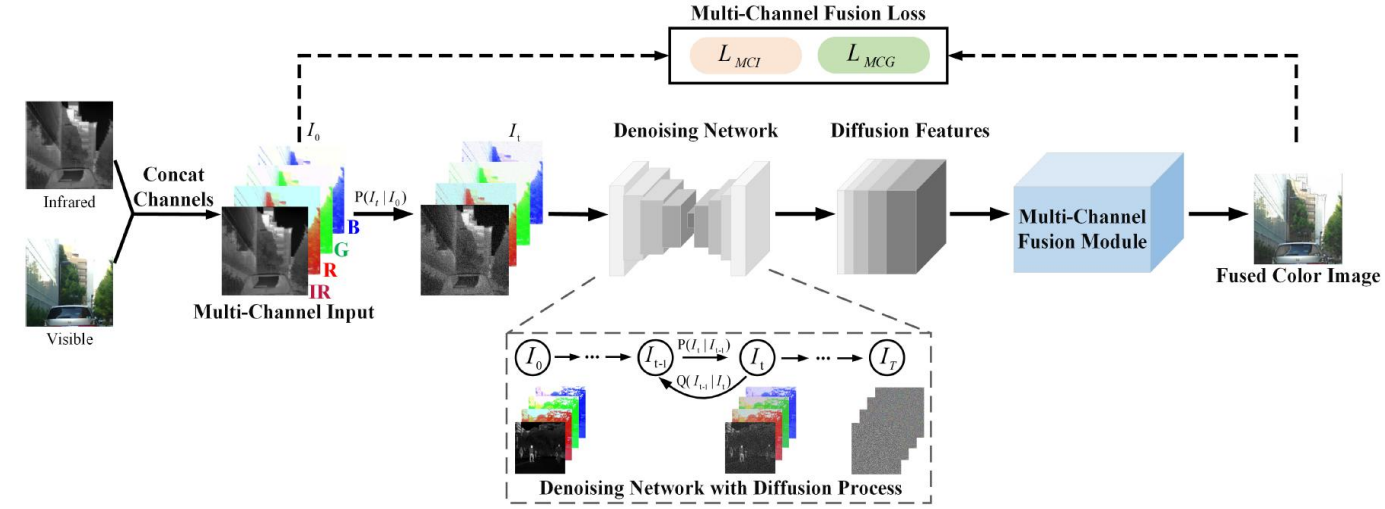
Three ways to fuse modalities

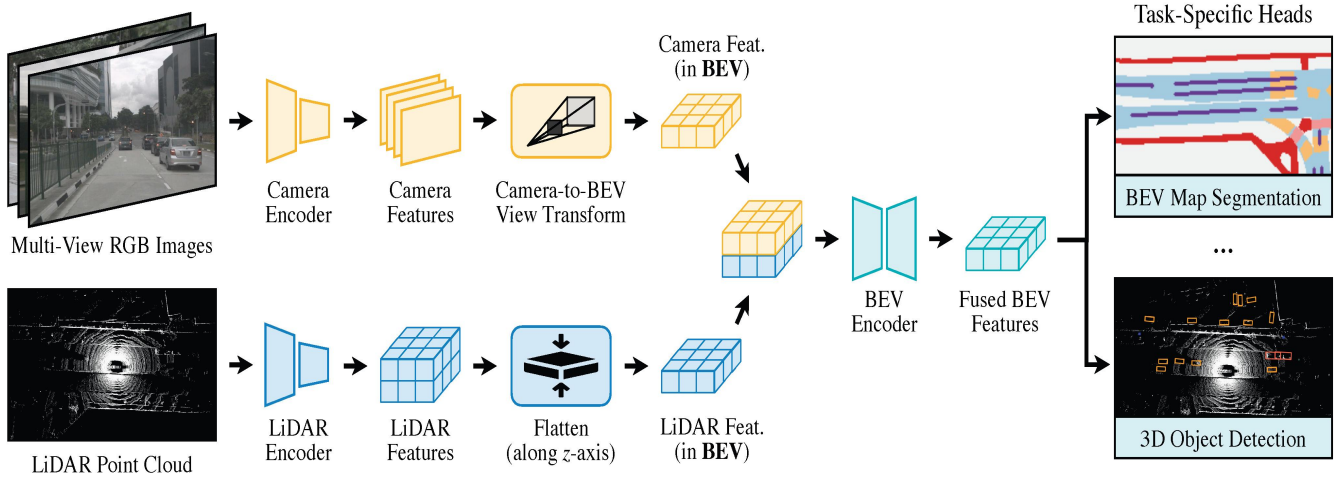
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Early Fusion (Data Fusion)
Sensor data is integrated at the input stage. The network learns joint features from the very beginning.

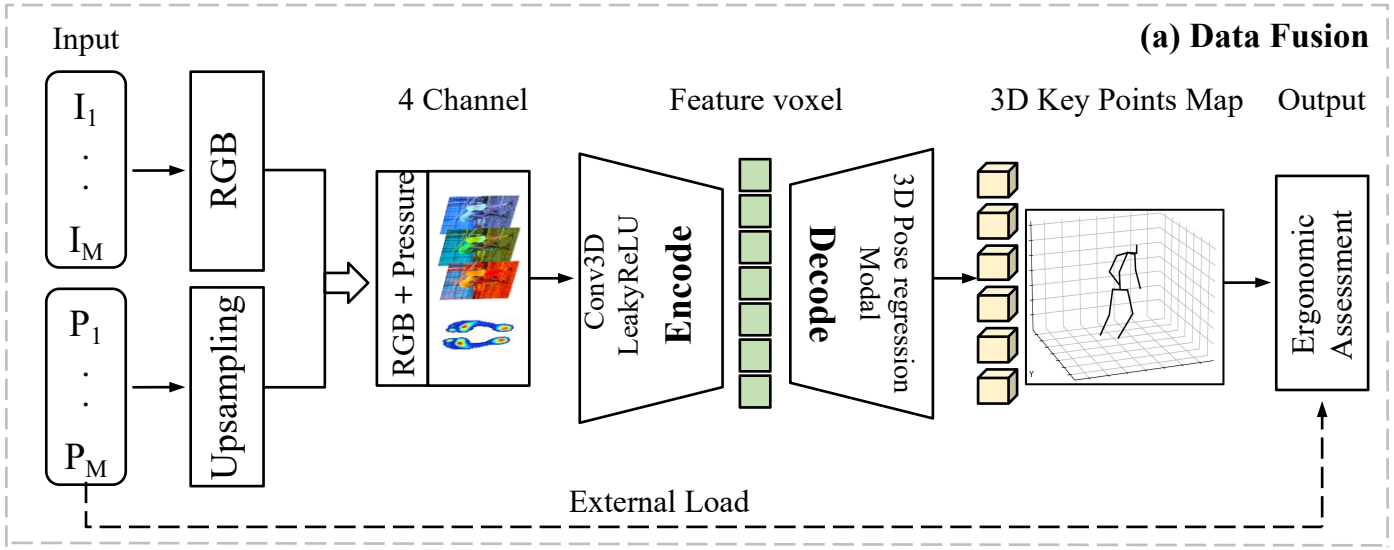
Intermediate Fusion (Feature Fusion)
Data from each sensor is encoded separately, and the resulting feature maps are then fused.

Late fusion (Decision Fusion)
Use different model to get results, and fused these results.





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Data Fusion: We used a two-dimensional interpolation method to create a pressure matrix for workers from 32-dimensional pressure insole data, thereby we standard the dimensions of foot pressure and visual data. We then combined the pressure data with the image channels and put them into the 4 channels multimodal model.

Fig. Research Methodology

Data Fusion

Input： Image data and pressure data

⇩

Create a pressure matrix and Up-sample
Multimodal data alignment

⇩

Data-level fusion of image and pressure

⇩

Multimodal pose estimation

Output： 3D pose data

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Feature Fusion: We used feature extractors to extract worker pose and pressure features, then connect and input them into the pose estimation module.

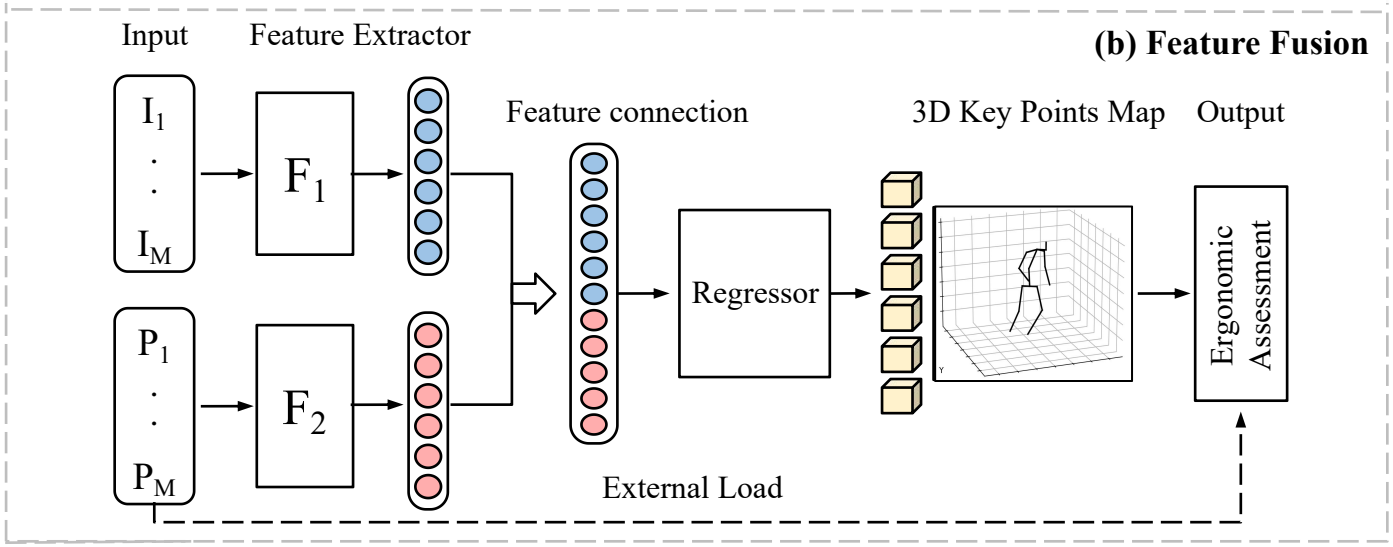


Fig . Research Methodology

Feature Fusion

Pressure & Image Feature Extraction

⇩

Multimodal data alignment

⇩

Feature-level fusion

Feature concatenation

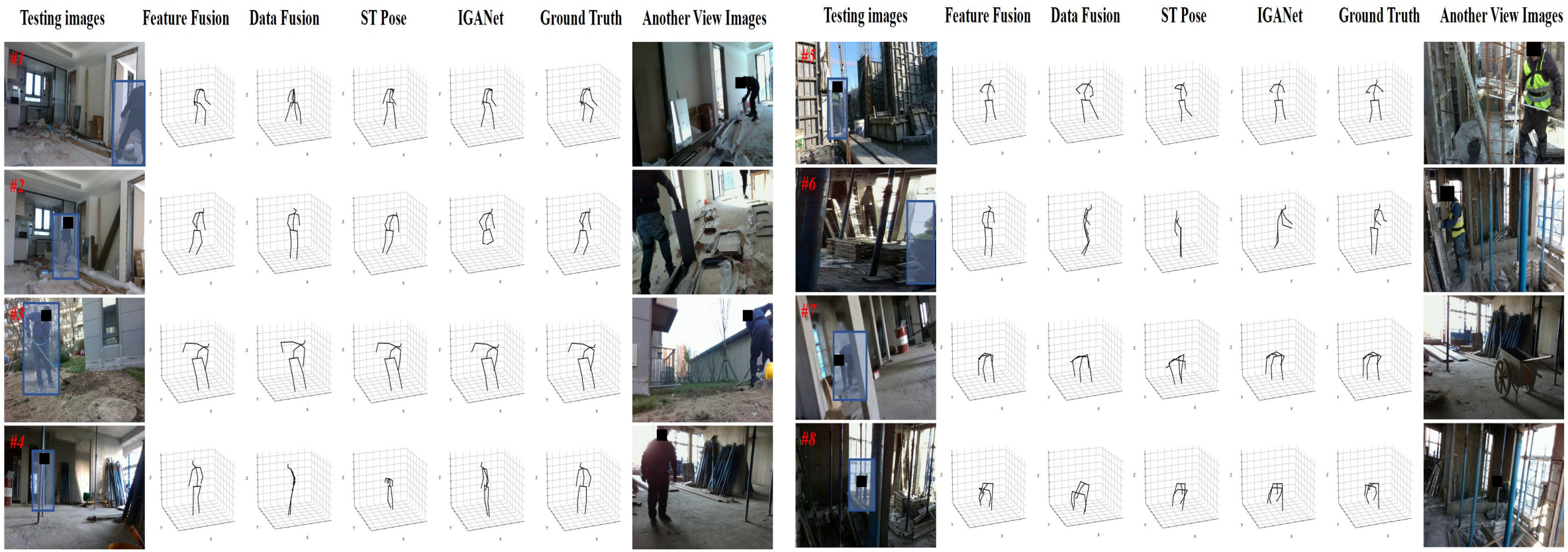
⇩

Multimodal pose estimation

Output：3D pose data

➢Results

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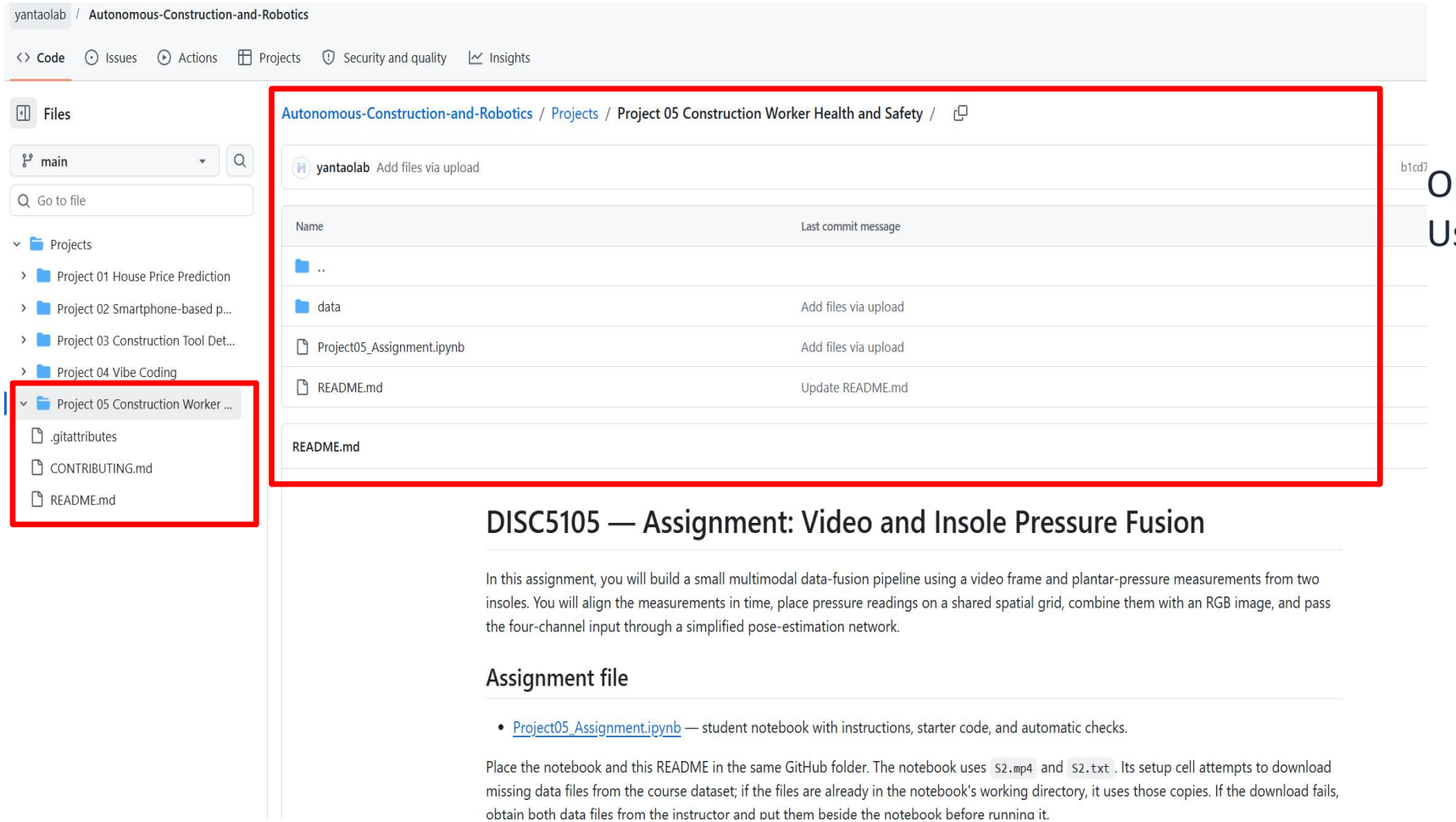


Examples 2 and 4: hidden or out-of-view joints show larger errors.

Assignment resources

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- The assignment notebook and dataset are available in the course GitHub repository.

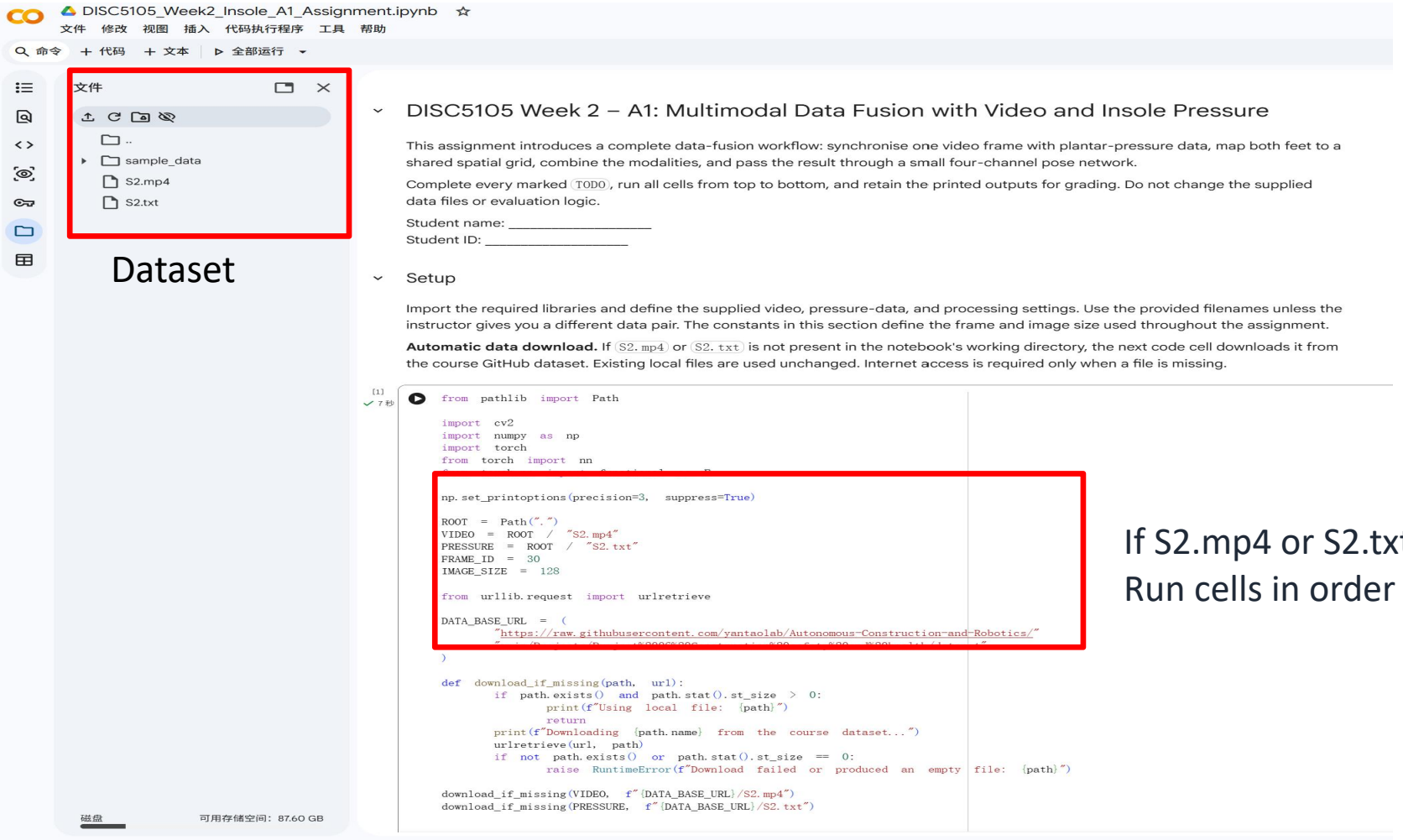


Open Project 05 in the repository.
Use the notebook and dataset in Colab.

Assignment workflow

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- Complete the assignment in Google Colab by filling in and running the provided code. The first cell is for downloading the dataset.



If S2.mp4 or S2.txt is missing, download it from github.
Run cells in order and keep the printed outputs.

Complete the assignment TODOs

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- Student tasks——data fusion
- Task 1 — Align video + pressure.
- Task 2 — Build RGB + pressure input.
- Task 3 — Build the pose model.
- Task 4 — Decode and check.

Edit the TODOs for Tasks 1–3.
Task 4 verifies the results; keep its evaluation code.

```
def align_and_build_pressure_map(times, left_pressures, right_pressures, frame_id, fps):
    """Align already-read TXT pressure data to one video frame and build a 1x13x13 map.

    Requirements:
    1. frame_time = frame_id / fps.
    2. Match pressure frames to video frames.
    3. Create a zero matrix with shape=(1, 13, 13).
    4. Write the row's 16 left-foot values into LEFT_SENSOR_POSITIONS.
    5. Write the 16 right-foot values into mirrored positions: right column = 12 - left column.
    6. Return pressure_map and matched_time.

    Returns
    -------
    pressure_map : np.ndarray, shape (1, 13, 13), dtype float32
    matched_time : float
    """
    frame_time = frame_id / fps

    # TODO 1 (one line): align frame_time with the already-read times and find
    # the pressure-row index with the smallest time difference. Hint: np.argmin(np.abs(...))
    nearest_index = None

    if nearest_index is None:
        raise NotImplementedError("Complete TODO 1: nearest_index")

    pressure_map = np.zeros((1, 13, 13), dtype=np.float32)
    for sensor_id, (row, col) in enumerate(LEFT_SENSOR_POSITIONS):
        # TODO 2 (one line): write the current left-foot sensor_id value to [0, row, col].
        pass
        # TODO 3 (one line): write the right-foot value to the mirrored position; right column = 12 - col.
        pass

    return pressure_map, float(times[nearest_index])
```

`TODO` lines

- Channel order is [R, G, B, P]
- Fused shape is (1, 4, 128, 128)

Expected outputs by task

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```
fps: 30.0 frame shape (BGR): (720, 1280, 3) frame time: 1.0

===== Pressure-map output =====
frame_time=1.000000s
matched_time=1.000000s
absolute_time_error=0.000000s
pressure_map shape: (1, 13, 13)
pressure_map[0] (13×13):
[[0.    0.    0.    5.5   0.    0.    0.    0.    0.    9.25  0.    0.    0.   ]
 [0.    0.    3.75  4.25  4.5   0.    0.    0.    6.5   3.25  9.5   0.    0.   ]
 [0.    2.5   2.    0.75  2.25  0.    0.    0.    1.5   0.25  4.5   2.    0.   ]
 [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.   ]
 [0.    0.    3.75  5.25  0.    0.    0.    0.    0.    1.75  1.25  0.    0.   ]
 [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.   ]
 [0.    0.    6.    5.25  0.    0.    0.    0.    0.    1.25  2.75  0.    0.   ]
 [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.   ]
 [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.   ]
 [0.    4.25  5.25  0.    0.    0.    0.    0.    0.    0.    1.25  1.    0.   ]
 [0.    0.    0.    1.75  0.    0.    0.    0.    0.    1.5   0.75  0.    0.   ]
 [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.   ]
 [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.   ]]
matched_pressure_index=10
```

TASK 1

```
===== Four-channel fusion output =====
pressure_scale=10.250000
fused shape: (1, 4, 128, 128)
RGB range: 0.019607843831181526 to 0.9764705896377563
pressure-channel range: 0.0 to 0.8989138603210449
one fused pixel [R,G,B,P]: [0.573 0.541 0.525 0.   ]
pressure channel 5x5 sampled preview:
[[0.    0.43  0.    0.631]
 [0.    0.019 0.    0.03 ]
 [0.    0.5   0.    0.158]
 [0.    0.112 0.    0.063]]
```

TASK 2

```
  (head): Conv2d(32, 17, kernel_size=(1, 1), stride=(1, 1))
)
input: (1, 4, 128, 128)
 stem: (1, 32, 32, 32)
 high: (1, 32, 32, 32)
 down: (1, 64, 16, 16)
  low: (1, 64, 16, 16)
 fuse: (1, 32, 16, 16)
 head: (1, 17, 32, 32)
low_up: (1, 32, 32, 32)
heatmaps: (17, 32, 32)
keypoints shape: (17, 3)
keypoint matrix [joint_id, x, y, confidence]:
[[ 0.    26.    46.     0.149]
 [ 1.   126.     6.    -0.004]
 [ 2.     2.    10.    -0.162]
 [ 3.    42.    90.    -0.031]
 [ 4.    38.    90.    -0.002]
 [ 5.    34.    30.     0.123]
 [ 6.   102.    14.     0.184]
 [ 7.    26.    66.     0.24 ]
 [ 8.     2.     2.     0.015]
 [ 9.    38.    46.     0.008]
 [10.    66.    98.     0.135]
 [11.   126.     6.    -0.063]
 [12.    50.    70.    -0.078]
 [13.    50.    70.     0.054]
 [14.    26.    58.     0.059]
 [15.   122.     2.     0.103]
 [16.    94.     2.    -0.065]]

===== Automatic evaluation =====
[PASS] pressure arrays
[PASS] nearest timestamp
[PASS] mirrored 13×13 map
[PASS] four-channel finite input
```

TASK 3–4

Submission checklist

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- Submit the completed Colab notebook or PDF with your name.
- Include Task 1 alignment and pressure-map evidence.
- Include Task 2 fused-tensor shape and range checks.
- Include Task 3 model printout and shape trace.
- Include Task 4 keypoint matrix and PASS/FAIL results.
- Add one limitation and one concrete improvement.
- Submit: Week2_A1_YourName.ipynb or Week2_A1_YourName.pdf

